Toward a metaplectic Jacquet–Langlands transfer, IV: Draft 1 Research and Verification Log

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Scope and provenance

This is the initial log for Part IV. The starting point was `mjl_part3_draft6.tex`, its revision log, the Part II planning memo, and the Kottwitz paper supplied in the conversation. The full final Part I manuscript was *not* available for a fresh, line-by-line independent cocycle verification in this drafting pass. References to Part I are therefore inputs to be rechecked when the cross-part referee report becomes available.

The title is aligned with the rest of the series: *Toward a metaplectic Jacquet–Langlands transfer, IV: Wild Local Correspondence and Global Trace Formula Comparison.*

New results and reductions

1. **Integral-character-lattice reduction.** For the unitary transform L_D constructed in Part III, integrality of coefficients in the irreducible character bases is equivalent to permutation of those bases, with a positive sign chosen by the identity-value normalization. The proof is elementary unitary linear algebra, not a proof of integrality.

2. **Finite-level computation for a fixed representation.** Because the fixed-central-character invariant functions are smooth on compact projective quotients, a single coefficient row can in principle be tested by a finite character calculation after passing to an appropriate sufficiently small level. The reduction uses the finite-expansion statement of Part III.

3. **Split diagonal-torus commutator radical.** Under the explicitly stated standard torus cocycle
$$\sigma_c(t,u) = (\det t, \det u)_n^c \prod_{i<j} (t_i, u_j)_n,$$
the commutator matrix is
$$A = -I_r + (1+2c)J_r.$$
When n is odd, $(n,r) = 1$, and $r - 1 + 2rc \equiv 0 \pmod{n}$, its radical on $(F^\times/F^{\times n})^r$ is the diagonal. Consequently a regular split diagonal element is good if and only if it belongs to $F^\times T^n$. This statement includes wild places, but *not* regular elements with nonsplit centralizers.

4. **Global arithmetic.** The almost-everywhere local-to-global result for $E^\times/(K^\times E^{\times n})$ from Part III is recalled with proof, but is explicitly separated from the still-open metaplectic local good-element and adelic measure questions.

Verification performed

- The diagonal torus commutator was calculated directly as the ratio of the two cocycles. The matrix has entries $A_{ii} = 2c$ and $A_{ij} = 1 + 2c$ for $i \neq j$.
- The row sum of A equals $r - 1 + 2rc$, matching the scalar-centrality congruence quoted in Part III. On the sum-zero summand of $(F^\times/F^{\times n})^r$, A acts as $-I$.
- The worked numerical case $(r, n, c) = (2, 3, 2)$ gives $A \equiv \left(\begin{smallmatrix} 1 & -1 \\ -1 & 1 \end{smallmatrix}\right) \pmod 3$.
- Kottwitz (1984), §§5, 7–9, is used for the arithmetic background only. Li's 2014 article is used only for the existence of a general invariant trace formula for certain adelic covers, not for an asserted identity in this problem.
- Zou's 2025 simple-type paper explicitly assumes $(p, n) = 1$, so it is not a published solution of the wild-place problem in the present draft.

Explicit unproved steps

Problem	What is still required
Part I consistency	Check the exact KP torus cocycle, inverse character conventions, and the star-map normalization
Wild local integrality	Prove integer character coefficients, or exhibit a counterexample to the proposed L_D
Nonsplit regular tori	Compute the commutator radical on $T_E(F)$ for every degree-r finite *etale* algebra E/F
Global transfer	Verify rational good-locus survival, identity and singular terms, and all Jacobian/measure conversions
Spectral separation	Find an actual global trace identity whose local coefficients are integers and that isolates representations

Questions for the cross-part referee

(Q1) Does the precise Part I torus cocycle restrict, up to a coboundary, to the convention displayed above? Does its scalar-center condition indeed match $r - 1 + 2rc \equiv 0 \pmod n$?

(Q2) Are the local $b_D = |n|_F^{-(r-1)/2}$ and $c_D = |n|_F^{-(r^2-1)/2}$ factors correct with precisely the stated fixed-central-character, quotient-probability Haar measures?

(Q3) Does the Part III smoothness proof really imply finite irreducible-character expansions in the nontrivial fixed-central- character projective setting?

(Q4) Are the Part II matching normalizations and the Part III trace pairing compatible when using inverse central characters for test functions and direct central characters for irreducibles?

(Q5) Does the global nth-power theorem use all required hypotheses, particularly $\mu_n \subset K$, $[E : K] = r$ and $(n, r) = 1$?

(Q6) Are any regular nonsplit torus good-element formulas in Flicker–Kazhdan or Kazhdan–Patterson sufficient to eliminate the remaining split-place task?

Completion assessment

Part IV, Draft 1 is a substantive beginning, not a completed proof of metaplectic Jacquet–Langlands. The split diagonal-torus result can be finalized after checking cocycle conventions. The wild-place integrality problem, general local good-locus theorem, global measure comparison, and spectral separation remain independent research obstacles. Their difficulty should not be described as merely editorial or a fixed number of revisions.